

Testing and Packaging

Testing

The system includes a pytest-based test suite to verify the correctness of core functionalities such as log parsing, searching, analytics, and exporting.

Unit tests are implemented in the following files:

- `test_parsing_searching.py`
- `test_analytics_reporting.py`
- `test_exporters.py`

These tests validate key features including:

- Template and generic log parsing
- Keyword search and filtering
- Log level classification
- Analytics calculations (counts, distributions)
- Export functionality (CSV/HTML)

To run all tests:

pytest

All tests must pass before packaging or deployment to ensure system reliability. The testing suite supports maintainability and helps detect regressions after code changes, aligning with the requirement for modular and testable code design.

Packaging

The application is packaged into a standalone executable using PyInstaller, enabling it to run without requiring a Python environment.

The packaging process uses the specification file:

- `LogAnalyzer.spec`

To build the executable:

python -m PyInstaller LogAnalyzer.spec --clean

The final executable is generated in the `dist/` directory:

dist/LogAnalyzer.exe

The packaged application was tested to ensure it runs correctly, including loading log files, performing analysis, and exporting results.